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Challenging critical thinking in education: new paradigms of artificial intelligence

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ABSTRACT

For many years, technological developments and innovations such as artificial intelligence (AI) has forced the education system to adapt and modernise. This new reality requires people to develop critical thinking (CT) skills to promote sustainable development and provide solutions to contemporary problems. However, traditional learning and teaching methods do not stimulate students' participation and motivation to develop and apply CT in the learning process. Given the situation in universities and the emergence of widely available AI tools, this study examines the progress of students' CT through field visits, seminars and workshops and determines the relationships between AI, education and CT. Two online and asynchronous questionnaires were administered before and after these activities, measuring 200 students' CT skills. The questions were related to students' perceived knowledge and reflections on the current economic, social and environmental landscape. The literature on the effects of AI on education was then reviewed. The results show that field visits, seminars and workshops positively affect CT development. In the post-questionnaire, students showed higher performance by reflecting on new items and justifying their personal stance in their answers, even though the questions entailed more complexity. These results suggest that teaching methodologies that place students at the centre of the learning process can stimulate active student behaviour. AI tools can complement CT, as long as governments and other economic agents direct their use towards ethical and transparent purposes.

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SUBJECTS
Artificial Intelligence; Technology in Education; Higher Education

1. Introduction

The current economic, social and environmental paradigm requires human capital with the thinking skills and knowledge needed to tackle the contemporary problems facing society. Therefore, university education must meet the new challenge of developing students by fostering contemporary knowledge and skills related to the global environment. Students should be at the core of the learning process, adopting a participatory role and eschewing passive behaviours. An effective learning process needs a dynamic attitude from individuals. However, context influences an individual's acquisition and application of knowledge (Almeida & Rodrigues Franco, 2011). Critical thinking (CT) entails reflective thinking, where an individual evaluates several statements and, after reflection, reaches a conclusion (Ennis, 1962). Whereas some scholars claim that CT depends on individual disposition, others argue that it emerges from a set of skills (Myrzatayeva et al., 2023). Thus, the academic research shows a lack of consensus on the implementation of CT. Although some authors claim that the application of CT can be extended to different fields (e.g. Halpern, 1998), others argue that it cannot be isolated from the applied field (e.g. Ennis & Weir, 1985).

These difficulties related to encouraging CT are compounded by the current demands of education that together challenge the effective learning process. Therefore, new learning alternatives are created

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to encourage students' active behaviour and participation. Students can thus feel motivated and actively engaged in the learning process. CT effectiveness is intrinsic to students' motivation (Facione, 1989). To gain an in-depth understanding of the relationship between motivation and decision making, Olivares et al. (2013) studied CT by applying the expectancy–value model, showing that the CT performance of students improves through motivational programmes. Similarly, Hajhosseini et al. (2016) found that discussion-based teaching enriches students' CT due to the exchange of agreements, disagreements, questions and answers. However, prior knowledge of a specific field is necessary to implement CT adequately (Stupple et al., 2017).

In the academic environment, the task of creating and developing learning alternatives that respond to current demands and can be effectively implemented is a challenge. This challenge has become even more complex with the emergence of new technologies and innovations that have disrupted and revolutionised the global paradigm in all fields of society, including education. Although CT is still a crucial skill of the 21st century (Spector & Ma, 2019), the importance of artificial intelligence (AI) and interest in this area of technology have increased in the last few years (Zawacki-Richter et al., 2019). The literature identifies some opportunities in the education system provided by AI, such as increasing students' curiosity, creativity (Dwivedi et al., 2023), and the quality of the learning process (Chen et al., 2020). However, questions about ethics and transparency (Pedro et al., 2019), privacy and security (Dwivedi et al., 2023) and the cost of maintenance and training (Kasneci et al., 2023) still remain.

Nowadays, achieving an effective learning process has become a challenge. The literature identifies possible problems encountered when developing CT skills at university (e.g. enhancing students' participation, motivation, interest and active behaviour). These problems are accentuated by the emergence and increasing availability of AI tools and other digital technologies. Given the lack of academic consensus on how to stimulate CT and the additional barriers added by AI tools, this paper addresses the research gap by pursuing two objectives. First, this paper examines attempts to encourage students' CT through activities such as external visits, seminars and workshops. Second, this paper provides an overview of the relationships between AI tools and education, leaving the open question of whether their effects are positive or negative. The external visits, seminars and workshops were undertaken in public and private institutions in the region of Valencia, Spain. The paper also examines the progress of students' CT in different groups of degrees. Specifically, the following research questions are addressed:

- Do practical and experiential activities (i.e. external visits, seminars and workshops) encourage students' CT skills?
- What is the progress of each student in terms of CT level?
- Are there differences between the CT performance of students from different undergraduate degrees?
- How are challenges and opportunities driven by AI tools affecting education and learning processes?

The activities and assessments were part of the World Economics subject taught at the Universitat Politècnica de València. This subject was taught in the degrees of (i) Business Administration and Management (BAM) and the double degrees of (ii) BAM and Computer Science, (iii) BAM and Telecommunications and (iv) BAM and Food Science and Technology. Two questionnaires administered before and after the activities (pre- and post-questionnaires, respectively) measured students' CT knowledge and skills. The CT question in the pre-questionnaire was 'Why do you think South Korea is immersed in a debt crisis?' In the post-questionnaire, the question was 'Do you think it is fair for companies to be taxed at a global level? Why?' The second part of the questionnaires dealt with specific world economy topics (i.e. regional competitiveness, big data and business intelligence, public bodies for promotion, and cooperatives and local development). In this part, students had to indicate their perceived knowledge about the topics and then answer some short questions on these topics. Both questionnaires were conducted online and asynchronously. Approximately 200 students participated in these field visits during the 2021/2022 academic year. The novelty of this research lies in the use of practical and experiential learning methods to connect directly with the business and social context surrounding the students to develop CT skills. In addition, the research emphasises the need to consider the opportunities

and challenges that AI or other digital technologies can raise in the academic environment to stimulate student engagement and in-depth knowledge acquisition.

The findings show that students' CT progress was positive. This positive trend was observed despite the greater complexity of CT on the post-questionnaire, which posed a more demanding question in terms of personal positioning and CT effort. Performance on the post-questionnaire improved from performance on the pre-questionnaire, with outstanding performance from double-degree students. This finding suggests that double-degree students may have higher CT skills because of the higher admission requirements for double degrees. Therefore, introducing learning activities that differ from traditional methods could stimulate student engagement and active learning. For instance, students could be encouraged to apply in-depth thought and reasoning to contemporary world problems. Such activities could provide them with fundamental tools to make decisions and understand their environment. Despite the growing global importance of AI, its effects on education remain unknown. In this case, the role of the government is crucial to direct the use of AI tools towards ethical and transparent purposes that complement the development of CT skills and knowledge.

This paper is structured as follows. Section 2 describes the characteristics of the Valencian business and innovation context and educational environment. Section 3 introduces the materials and methods used to conduct the research. Section 4 presents the results of the analysis and the effects of participatory activities on students' CT performance. Section 5 discusses the challenges and opportunities of integrating AI tools into the education system. Finally, Section 6 concludes and proposes future research possibilities in response to the paper's limitations.

2. The business, innovation and educational context in the region of Valencia

The region of Valencia is a Spanish region on the Mediterranean coast. It has the fourth most small and medium enterprises (SMEs) of all Spanish regions. Catalonia accounts for 18.4% of Spain's SMEs, followed by Andalusia (16.2%), Madrid (15.7%) and the region of Valencia (11%). Together, they are home to 61% of the SMEs in Spain (Dirección General de Estrategia Industrial y de la Pequeña y Mediana Empresa, 2024a), where 99.8% of all companies are considered SMEs (Dirección General de Estrategia Industrial y de la Pequeña y Mediana Empresa, 2024b). The region of Valencia has 352,955 companies, of which 82% operate in the service sector, 11.7% in the construction sector and 6.3% in the industrial sector (Generalitat Valenciana, 2024).

Higher education in Valencia is provided by six universities: two public and four private. The public universities are the Universitat Politècnica de València and the *Universitat de València*. The students of the Universitat Politècnica de València have the opportunity to attend field visits and workshops at some of the organisations that make up the Valencian business and economic ecosystem. Examples include the Business Confederation of the Region of Valencia (*Confederación Empresarial de la Comunidad Valenciana*, CEV), the Business Competitiveness Institute of the Region of Valencia (*Instituto Valenciano de Competitividad Empresarial*, IVACE), ZEUS Vision, Sesame and the Alto Palancia Olive Oil Cooperative. CEV represents and defends the interests of firms in the region of Valencia, promoting competitiveness, innovation, economic development and social progress by engaging with public and private entities (CEV, 2024). IVACE is a public entity run by the *Generalitat Valenciana* (the regional government) to enhance competitiveness, focusing on areas such as innovation, financial support, industrial infrastructure, collaboration and business internationalisation (IVACE, 2024). Zeus Vision, which specialises in business intelligence, data visualisation and digital transformation, and Sesame, which focuses on work team management software, are innovative private companies. Finally, the Alto Palancia Olive Oil Cooperative is an association of local cooperatives dedicated to producing high-quality extra virgin olive oil.

The Universitat Politècnica de València is a public university known for its dynamism and innovation in research and education. It has strong ties with its community and seeks to have a major international presence. The task of transferring and connecting students' learning with the Valencian environment is crucial because they have to make decisions and act within that environment. The Universitat Politècnica de València was founded more than half a century ago. It currently has approximately 28,000 students, 2500 researchers and professors and 1500 administration and services staff spread over three campuses in Alcoi, Gandia and València. Although it has always specialised in engineering and

technology, the university has strived to meet society's needs by including other related degrees, such as public and business administration (UPV., 2024).

3. Materials and methods

Many scholars have studied CT, attempting to set some basis for understanding and measuring CT in the educational sphere. Halpern (1998) suggested that, when teaching and providing students with skills to think critically, the goal should be not only to focus on students' success in that particular skill but also to ensure that they can recognise the kinds of situations where they can apply that skill. How can this goal be achieved in an educational programme? Franco et al. (2014) assessed CT in terms of academic performance and concluded that the higher the academic performance is, the higher the CT development is. However, they questioned whether this CT development is triggered purely by academic learning or by other factors. Identifying which elements of the learning process influence the development of CT remains an open question.

This study aims to (i) encourage students' CT and explore students' CT progress and (ii) determine the possible challenges and effects of technological development (e.g. AI) on global educational systems. Students enhanced their CT by attending external or field visits, workshops and seminars with experts. These activities were undertaken in public and private institutions in the region of Valencia such as the Business Confederation of the Region of Valencia (*Confederación Empresarial de la Comunidad Valenciana*, CEV), the Business Competitiveness Institute of the Region of Valencia (*Instituto Valenciano de Competitividad Empresarial*, IVACE), ZEUS Vision, Sesame and the Alto Palancia Olive Oil Cooperative. The activities and their assessment were part of the World Economics subject taught at the Universitat Politècnica de València. The degrees that included this subject were (i) BAM and the double degrees of (ii) BAM and Computer Science, (iii) BAM and Telecommunications and (iv) BAM and Food Science and Technology. Two questionnaires administered before and after the activities (pre- and post-questionnaires, respectively) measured students' CT knowledge and skills. Part of students' final grade for the subject (10%) was awarded for answering both questionnaires and attending at least two visits, seminars or workshops. This incentive ensured student participation in the activities and thoughtful responses to the questionnaires. Approximately 200 students participated in these field visits. The data were processed anonymously and aggregated. This study is part of a teaching innovation project entitled 'The role of artificial intelligence in the transformation of the university system: active learning and competence development' (in Spanish, '*El papel de la inteligencia artificial en la transformación del sistema universitario: aprendizaje activo y desarrollo de competencias*'; PIME/23-24/398). This project is part of an educational innovation and improvement programme funded by the Institute of Education Sciences (in Spanish, ICE) of the Universitat Politècnica de València.

Designing and applying the two questionnaires to measure the activities' impact on the development of students' CT skills enabled assessment of the capacity for critical reflection on topics related to the current economic, social and environmental context. The two questionnaires were administered online and asynchronously. Each student had one week to complete them. First, students were asked about their perceived knowledge about specific topics (i.e. regional competitiveness, big data and business intelligence, public bodies for promotion, and cooperatives and local development). The students responded on a scale ranging from very low to very high. Other studies (e.g. Roohr & Burkander, 2020) have also employed students' perceived knowledge of a topic or subject to evaluate CT. Then, students were asked some short questions on these specific topics. Combining multiple-choice and open-ended questions can benefit learning because students have to justify their initial answers to the multiple-choice questions (Kleemola et al., 2022). To measure CT competence, students were asked to comment on a news item, which differed in each applied questionnaire, related to the study topics. According to Kleemola et al. (2022), open-ended questions can enhance holistic CT processes. The questionnaires were marked based on a list of keywords that should appear in the responses and the reasoning and arguments used by students considering the information provided by the news item.

The CT question in the first questionnaire was 'Why do you think South Korea is immersed in a debt crisis?' In the second questionnaire, the question was 'Do you think it is fair for companies to be taxed at a global level? Why?' The coding of answers consisted of assigning a value to the students' answers

Table 1. Questions and coding.

Question		Coding of answers in terms of CT development
Read and comment on the following news item:	**Pre-questionnaire** Why do you think South Korea is immersed in a debt crisis? **Post-questionnaire** Do you think it is fair for companies to be taxed at a global level? Why?	• 0: very poorly developed • 0.25: poorly developed • 0.5: moderately developed • 0.75: well-developed • 1: highly developed

based on a scale ranging from 0 to 1. The following marks were given to answers: (i) 0 for very poorly developed answers, (ii) 0.25 for poorly developed answers, (iii) 0.5 for moderately developed answers, (iii) 0.75 for well-developed answers and (iv) 1 for highly developed answers. The post-questionnaire included 12 items with questions related to the student's satisfaction with the visits, seminars and workshops. These questions were coded on the following Likert scale: (1) *strongly disagree*, (2) *disagree*, (3) *neither agree nor disagree*, (4) *agree* and (5) *strongly agree*. Table 1 summarises the questions and the coding process. It shows the questions that students answered and how their answers were coded. The sample and methodology had potential biases related to conducting activities in a specific context, students' knowledge background and researcher subjectivity in coding. These limitations are further explained in the section on conclusions.

Students gave a variety of answers to the CT questions. Regarding the pre-questionnaire, one of the students was of the opinion that '*one of the main causes of South Korea's debt crisis is the global pandemic. It has caused prices to rise, making a large part of the population go into debt to make ends meet. And due to this negative situation, many indebted people are finding it impossible to repay their debts*'. This answer was considered highly developed and was thus coded as 1. Another student replied that the main reason for the South Korean debt crisis was '*because the South Korean economy was transformed into a high-income and, therefore, also high-cost economy; added to the corruption that was taking place, South Koreans ended up living beyond their means, and this could cause them to go into debt*'. This answer was coded as 0.75 and was considered well-developed. In contrast, the answers '*because the interest rate has gone up and there is a high youth unemployment rate*' and '*I do not know*' were coded as 0.5 and 0, respectively.

Regarding the post-questionnaires, some examples that were coded as 1 (highly developed) were '*I do think that establishing a worldwide tax rate that affects companies is fair, as many of them declare taxes in countries other than the one in which they are operating, which creates great advantages for them, but great disadvantages for the country in which they operate but do not declare*' and '*In my opinion, the same tax rate should not be set for all countries. I think a tax rate should be set according to the organisation's size. In other words, I would set one tax rate for SMEs and another for large companies. In addition, I would set a higher tax rate for companies that use hardly any of their home country's national resources*'. The answers '*I think it is fair because it ensures greater equity in terms of international tax arrangements*' and '*I do not think it is fair. Maybe for those multinational companies that operate globally, it is fair, but not for the rest*' were considered well- and moderately developed and were thus coded as 0.75 and 0.5, respectively. The answer '*I do not know*' was again coded as 0.

4. Results

Students' CT progress was positive, even though the post-questionnaire introduced greater CT complexity because students were required not only to reflect on the new item but also to take a personal stance in their answer. In the first questionnaire, 68.9% of the answers were considered to have high or very high levels of CT. After attending the field visits, 92.5% of students showed high or very high levels of CT. Therefore, the CT level increased from 68.9% (pre-questionnaire) to 92.5% (post-questionnaire). The answers to the second questionnaire reveal that, when students were asked about the need for a global tax rate for companies, they adopted different positions, varying from complete agreement to disagreement. Regardless of their position, in most cases (80.7%), they argued their answers effectively.

Figure 1 compares the percentage of students with different CT levels in the pre- and post-questionnaires. In the pre-questionnaire, students had different levels of CT, with most answers coded as medium, high and very high CT levels. However, the post-questionnaire values show that most students (80%) gave

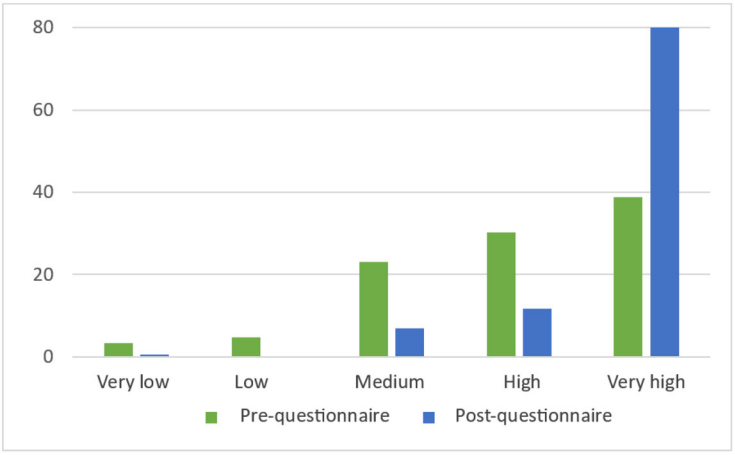


Figure 1. Students' CT levels comparing pre- and post-questionnaire results.

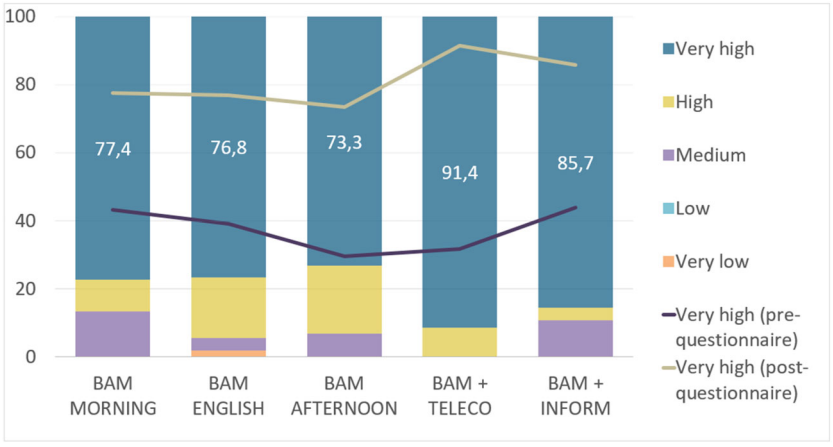


Figure 2. Level of CT by degree (%) in the post-questionnaire.

highly developed answers, applying very high levels of CT. The percentage change for different levels of CT was also calculated. The lowest values of CT development decreased more than the highest levels, which increased. The lowest CT levels (very low and low development) decreased by more than 80% from the pre- to the post-questionnaire. Medium and high levels decreased by between 60% and 70%. In contrast, very high CT levels increased by more than 100% from the pre- to the post-questionnaire. These findings confirm the progress in students' CT, suggesting that CT skills can be enhanced and developed through more dynamic and participatory activities such as field visits, seminars and workshops.

Figure 2 shows the percentage of CT development levels for each degree in the post-questionnaire. The number of highly developed answers (very high CT level) increased in all degrees. Although the students from all degrees performed better in the post-questionnaire, the answers in the double degrees were particularly well developed, reflecting higher CT levels. All the answers for the degree of BAM and Telecommunications reflected high or very high CT levels. BAM and Telecommunications students increased their CT levels from 31.6% in the pre-questionnaire to 91.3% in the post-questionnaire. In the pre-questionnaire, students from more technical degrees achieved more uniform results than BAM students. However, in the post-questionnaire, the progress in CT was positive for all degrees, and both BAM groups of students achieved similarly high percentages in CT level.

5. Artificial intelligence (AI): challenges and threats in education

Human development faces new challenges due to the growing importance of information and communication technologies (ICTs) and their integration in all spheres of life (Kouzov, 2019). The implementation of new technologies and innovations is revolutionising education, triggering a disruptive change in

the learning and teaching process (Spector & Ma, 2019). AI is now in the public eye worldwide. According to Chen et al. (2020), studies should focus not only on AI but also on the developments and innovations that result in human intelligence computers or other technologies. These innovations are characterised by decision-making capabilities, learning adaptability and cognitive abilities. Creating an ecosystem of flourishing innovators and advanced infrastructures is essential for successful AI. These ecosystems are complex because they combine innovation, knowledge, regulation and business relationships (Pedro et al., 2019). In recent years, the number of applications of AI in education has increased, placing AI in the spotlight (Zawacki-Richter et al., 2019). Therefore, it is paramount to consider how AI can influence higher education in the design and implementation of learning strategies that ensure the success of students' development, motivation, interest and participation. This section presents some of the challenges and threats that AI poses in higher education.

The literature refers to AI education in situations where learning and teaching contents and elements stimulate problem-solving, decision-making and creativity skills by employing AI innovation and technologies (Shin, 2021). However, the application of AI in education has been shown to have positive and negative effects on the education system (Kasneci et al., 2023). AI improves the quality, efficiency and effectiveness of the learning and teaching process, ensuring deep learning (Chen et al., 2020). Grájeda et al. (2024) showed that students' academic experience is enriched by AI tools, which enhance productivity, creativity and understanding. From the students' perspective, AI tools (e.g. ChatGPT) may be a source of inspiration and creative ideas (Dwivedi et al., 2023), stimulating students' CT. Therefore, applying AI tools as innovative learning methodologies can stimulate key skills for students. This idea may be linked to the study's findings because expert seminars and field visits could also be introduced as new learning methodologies that encourage students' participation, deep knowledge acquisition, motivation and improvement of CT skills. Moreover, AI tools make information immediately available, which may facilitate the research process and encourage students' curiosity about reviewing their own questions and answers (Dwivedi et al., 2023).

From the perspective of professors and lecturers, AI tools can also lead to other important positive changes in education. These tools can enable personalised learning for students by facilitating the creation of class plans and activities that are inclusive of all types of students (Kasneci et al., 2023). Professors and lecturers could use AI tools to perform research or writing duties more effectively and efficiently. Some of these duties may include providing feedback to students or producing work for seminars. Moreover, AI can provide professors and lecturers with explanations, overviews and resources related to new learning and teaching materials or methodologies (Kasneci et al., 2023). Thus, they can invest their time and effort in more complex research or education tasks that require CT and creativity (Pedro et al., 2019).

Dwivedi et al. (2023) offer an overview of AI tools such as ChatGPT, according to the opinions of experts from different fields. As claimed by several scholars, the authors argue that AI enhances education opportunities by creating new educational material, providing personalised feedback and language learning support and automating administrative tasks. Their study also shows that online education should be encouraged in the educational system because AI tools can provide students with more resources and instruments that improve online learning by making learning experiences more participatory and exciting. Applying these findings to the current study, activities such as expert seminars, workshops and field visits transform learning experiences using face-to-face student participation as opposed to online resources, where students directly connect with the environment and the economic and social context. Online and face-to-face learning experiences that depart from traditional learning methodologies can have crucial positive effects on education. Moreover, professors and lecturers could tutor students individually by addressing specific questions. They could thus ensure students' comprehension (Dwivedi et al., 2023).

Despite the benefits of AI in education, some key challenges should be addressed. AI can create technological, social and economic gaps between developed and developing countries because not all countries have sufficient resources and technological infrastructures to implement AI successfully in learning and teaching. Therefore, ensuring AI equity and inclusion in education may create problems at the national level (Pedro et al., 2019). Moreover, professors and lecturers usually lack training and instruction in AI technology and tools to support student learning. Hence, adopting AI technology may be problematic (Chiu & Chai, 2020; Kopcha et al., 2020). Moreover, in educational institutions, AI tools and methods are not always affordable, given the high cost and time required for their implementation (Welham, 2008; Zawacki-Richter et al., 2019).

ChatGPT is currently one of the most popular AI tools. However, it is an expert in generating sentences that sound convincing but are meaningless and even sometimes false (Mollick, 2022; Dwivedi et al., 2023). The output from ChatGPT may be incorrect or misleading because the algorithms and data it uses may be inaccurate. Therefore, students should always check the answers from this tool and verify this information with other sources before submitting any assignment. According to Crowe et al. (2017), the adverse impacts of AI on education and learning are clear. They argue that AI jeopardises academic integrity and encourages dishonesty (Chen et al., 2020). This reasoning can also be applied to scientific production. ChatGPT leads to an increase in the number of unoriginal papers that usually contain unnoticed errors and sound convincing. This AI tool spreads misinformation, distorts scientific facts and encourages plagiarism (Dwivedi et al., 2023).

Kasneci et al. (2023) identified the major education-related challenges of AI. Answers from AI tools may lead to copyright issues because some paragraphs or sentences may be copied from a particular source of information. The effort when using AI tools encourages overconfidence in the outcome, increasing individuals' laziness and decreasing interest in developing their own research or conclusions. This feature of AI may limit the development of creativity, problem-solving and CT skills. Moreover, distinguishing human- and AI-generated content is difficult. AI tools can generate text to resemble what humans would write, although the information it contains may be unverified and may thus contradict actual knowledge. Students might accept this misleading and false information without questioning its validity. The implementation of AI in education also raises privacy and security problems because ChatGPT and other AI tools have access to sensitive personal data (Dwivedi et al., 2023), which may include access to unauthorised data or data breaches (Kasneci et al., 2023). Therefore, ethical and transparency issues are relevant. Hence, creating a policy on ethics, transparency and security is important.

The task of designing and implementing a comprehensive policy that simultaneously addresses sustainable development and AI is complex because it entails aligning numerous institutions and factors that are directly or indirectly involved in the process (Pedro et al., 2019). However, such policies are necessary and practical. Adequate implementation could mitigate AI's negative effects and promote positive ones. For example, the OECD. (2024a) has tried to bring together country policies through initiatives that address multidisciplinary issues. Since 2016, the OECD has run several activities and programmes to collect information on AI, establishing a series of AI principles in 2019 (updated in 2024). These principles provide a framework for developing reliable AI, offering recommendations for AI policy-making and classifying AI risks and future possibilities in several dimensions such as R&D, inclusive AI ecosystem, governance and policy environment, human capital capacity and global cooperation (OECD., 2024b). Filgueiras (2022) analysed the relationship between political regimes and AI development, reflecting that developing countries' authoritarian governments are more productive in AI and find fewer obstacles to AI production. For example, China has a policy that combines geopolitical, fiscal, ethical and legal objectives, whereas Russia's policy extends Internet governance and individual and business network access through extensive information flow control and monitoring (Filgueiras, 2022).

These examples show that, despite different country characteristics, cultures and interests, it is essential to collaborate and include everyone in AI development. This development should be directly connected to the importance of establishing trustworthy guidelines for ethical and transparent AI use. Otherwise, inequalities will increase due to the technological, social and economic gap that emerges from considering each country's own interests separately. Such an approach would jeopardise the achievement of sustainable development and global well-being. Benefits derived from AI are not equally distributed among countries, with developing countries having limited access to technologies, research and infrastructure (Mannuru et al., 2023). Mannuru et al. (2023) claimed that the contribution of AI to inclusive development calls for the necessary infrastructure and support to reduce country inequalities.

Education improves and develops human intelligence. CT has become a fundamental skill for the 21st century (creativity, CT, collaboration and communication). However, investment in education is lower than investment in AI (Spector & Ma, 2019). A worrying aspect about using AI tools is the risk of deterioration of language expression and independent thinking skills, which are not trained and developed when these tools are employed. AI is not a substitute for problem-solving, creativity and CT skills. Instead, it provides fast answers to questions (Dwivedi et al., 2023). According to Willingham (2019), AI and other emerging technologies have increased the importance of CT around the world. Despite the retractors and negative

perception of AI's effect on education and CT, AI was never meant to replace human intelligence and CT. It was created as a complementary tool to meet society's needs without triggering social or moral problems (Kouzov, 2019). Educational systems could adopt AI as a teaching tool focused on CT, taking advantage of its benefits for curricula and learning environments (Kasneci et al., 2023).

AI can generate dynamic learning experiences, where students feel connected with technology and the current disruptive environment. Such an environment enhances CT, creativity and other skills. However, the advantages and disadvantages of implementing AI tools in education should be addressed to ensure students' effective and in-depth learning. Combining online and face-to-face education can enrich the learning process because of the interactive and stimulating learning environment, where students are placed at the centre of skill development and learning. One example of combining online learning (AI tools) and face-to-face educational activities could include linking AI-based platforms (e.g. Edpuzzle) with flipped classroom learning, with student performance tracked by AI before class assignments. Professors and lecturers could thus identify questions and adjust classes according to students' needs. Finally, the activity could finish with the participation of students in the class explanation. Other examples involve employing AI tools to complement creativity tasks in project- or problem-based learning.

6. Conclusions

This paper shows that designing and implementing new learning activities in educational ecosystems that diverge from traditional methods could encourage student participation and active learning. Students could thus be encouraged to apply in-depth CT and reasoning to the world's current problems. In turn, such an approach could offer students the skills and tools needed to understand their environment and make decisions that promote sustainable development. Professors and educators are responsible for training and developing students' CT (Dwivedi et al., 2023). AI is receiving growing attention worldwide (Spector & Ma, 2019). However, its effects on education are unknown, and the lines of action and management that deal with AI have not yet been established. Hence, there are many open questions about the overall effect of AI on the education system and, more specifically, on CT, which remains an essential skill to be built during the learning process (Willingham, 2019).

AI has been at the forefront of the media for several years. The launch of ChatGPT revolutionised the news and social networks, bringing both the positive and negative aspects of AI into the spotlight (García-Peñalvo, 2023). This high level of media attention on ChatGPT has made software providers such as Microsoft and Outlook increase their interest in these types of AI tools by integrating them into PowerPoint and Word. Therefore, although some students are still unfamiliar with AI tools, their inclusion in globally recognised software will extend their use (Dwivedi et al., 2023). However, the question is how AI will impact the current education system and thus the learning and teaching process that is supposed to encourage CT. Numerous studies have detailed the positive and negative effects of AI (e.g. Pedro et al., 2019; Zawacki-Richter et al., 2019; Chen et al., 2020; Dwivedi et al., 2023; Kasneci et al., 2023).

Although it has been suggested that AI tools can inspire students and stimulate creative ideas that drive CT, they make information rapidly available for all individuals (Dwivedi et al., 2023). This situation may lead students to accept all information as valid, without questioning it and perhaps without reviewing the solutions provided by these tools. Such a scenario compromises academic integrity and encourages dishonesty (Chen et al., 2020) because individuals can directly copy results without prior reasoning or thinking. It also leads to plagiarism and copyright issues (Kasneci et al., 2023). Students may trust the AI tool and may not push themselves to apply CT to perform classroom activities, even when making important decisions. Therefore, thinking and creative skills may deteriorate, along with other such skills, leaving future human intellectual capacity in a situation of uncertainty regarding problem-solving and decision-making potential. If the use of these AI tools continues to increase, many individuals will be reluctant to apply an intellectual effort to undertake any activity because it is simply easier to use AI tools, which, in a few seconds, provide an answer that makes logical sense at first glance, especially if the user is unfamiliar with the topic of interest (Mollick, 2022).

In this context, professors and lecturers play a crucial role in students' efficient and effective use of AI tools. Professors and lecturers have the power to emphasise CT skills and explain how AI can add value. They can teach students how to ask themselves questions and find the answers (Dwivedi et al., 2023).

For example, Darwin et al. (2024) argue that students recognise AI's usefulness in enhancing aspects of CT such as theory scrutiny and academic research. At university, professors and lecturers can teach students the different ways ChatGPT can help them to improve language skills, create concepts and structure their writing, as well as determining appropriate sources for writing an essay (Qadir, 2022). Other examples of AI applications in education are the writing of traditional essays and oral presentations, the application of a concept to a specific situation, and in-class exams or written questions that require high comprehension of an individual's own experiences and relationship with knowledge (Dwivedi et al., 2023). However, not all professors and lecturers understand and are prepared to implement AI in education. They first need to acquire digital skills to employ AI tools as teaching methods (Pedro et al., 2019).

An open question is how to apply these new AI technologies and innovations in the education system so that students avoid lazy and passive behaviours and instead use these tools to improve their CT when addressing contemporary problems and sustainable development. The future of AI in education and other areas of society is still uncertain. It is unknown whether its overall effect on humankind will be positive or negative. Will we have to wait to see these future effects? Could human beings intervene in a way that will channel these technologies to foster and complement CT? Policymakers can address these concerns and problems by implementing policies, regulations and other measures to ensure social trust, privacy and transparency in the application of AI tools in the education ecosystem (Pedro et al., 2019). The role of the government may be central to guaranteeing AI's success and adaptability in the learning and teaching process in a way that boosts the participation, action, and CT behaviour of students and other agents of the education system. Therefore, educators and policymakers should consider four key aspects when implementing new programmes or learning activities: (i) accessibility, (ii) privacy, ethics and transparency, (iii) human development and (iv) equity and inclusion. To ensure effective and in-depth learning, it is paramount to ensure that AI tools are accessible and available for all students to give even those from small communities the required infrastructure. This aspect is related to students' inclusion. Establishing guidelines and policies that ensure students' and others' privacy is essential to guarantee ethics and transparency. Finally, AI tools will not succeed unless individuals continuously engage in training and development of AI skills.

This research is not exempt from limitations. First, the findings refer to external field visits, seminars and workshops in a specific and limited context. Therefore, findings in other contexts could differ. In addition, although the participating students had some engineering knowledge and skills (i.e. telecommunications and computer science), most were from a predominantly business administration and management background. The results could also vary depending on the knowledge area in which a similar analysis is conducted. Moreover, differences in performance improvement between control and treatment groups could not be tested because it was impossible to prevent students from participating in the activities. Other limitations relate to researcher bias because of the subjective nature of answer coding. Future research possibilities include considering a higher number of students from different degrees in the sample to identify not only the global impact of active learning methodologies on CT but also any possible differences in CT levels by degree. Finally, AI tools could be introduced in classroom activities to analyse their impact on students' CT levels.

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